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A COMPARATIVE STUDY OF FRICTION ESTIMATION AND COMPENSATION USING EXTENDED, ITERATED, HYBRID, AND UNSCENTED KALMAN FILTERS

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ABSTRACT

Transparency is a key performance evaluation criterion for haptic devices, which describes how realistically the haptic force/torque feedback is mimicked from a virtual environment or in case of master-slave haptic device. Transparency in haptic devices is affected by disturbance forces like friction between moving parts. An accurate estimate of friction forces for observer based compensation requires estimation techniques, which are computationally efficient and gives reduced error between measured and estimated friction. In this work different estimation techniques based on Kalman filter, such as Extended Kalman filter (EKF), Iterated Extended Kalman filter (IEKF), Hybrid extended Kalman filter (HEKF) and Unscented Kalman filter (UKF) are investigated with the purpose to find which estimation technique that gives the most efficient and realistic compensation using online estimation. The friction observer is based on a newly developed friction smooth generalized Maxwell slip model (S-GMS). Each studied estimation technique is demonstrated by numerical and experimental simulation of sinusoidal position tracking experiments. The performances of the system are quantified with the normalized root mean-square error (NRMSE) and the computation time. The results from comparative analyses suggest that friction estimation and compensation based on Iterated Extended Kalman filter both gives a reduced tracking error and computational advantages compared to EKF, HEKF, UKF, as well as with no friction compensation.

1 INTRODUCTION

Friction forces limit the transparency of force/torque feedback mimicked by haptic devices from a virtual environment or in case of master-slave haptic devices. Transparency mean, that the user should neither feel the force/torque from dynamics of the device, nor the friction in

the moving parts or back-EMF (electromotive force) of the motor, either in free space or in collision. When an operator moves the tool center point (TCP) of the device in free space or in collision with an object in the virtual environment, friction in the joints of the device resists the operator's motion. The resistant friction forces contribute to the operator's fatigue [1], and the force/torque feedback is not perceived as realistic.

The performance of a haptic device is often deteriorated by friction. Friction can induce unwanted effects, such as steady-state errors and limit cycling. The usage of a haptic device, involves small displacements at low and reversal velocities. In particular, many physical phenomena, such as stiction and pre-sliding can result in stick-slip motion that has a big influence on device performance and stability. High quality haptic interaction requires compensation of frictional effects, which relies on a friction model that enables accurate prediction and real-time compensation of the friction forces.

The 6-DOF Ares haptic device [2] based on Stewart platform suffers from friction. This device is developed to create a sense of touch to the user in a virtual environment. Due to the presence of friction, however, the device does not feel realistic enough. Especially when motion changes direction, the user feels an undesired reversal bump.

2 RELATED WORK

In practice, various *ad hoc* compensation strategies are addressed in the literature, such as non-model-based and model-based friction compensation. For non-model-based friction compensation, no model of the friction force is needed. Examples of this type of compensation are disturbance observer base compensation as in [3], [4].

Model-based friction compensation uses a friction model to predict friction force and compensate for it. The friction

model must include all major friction characteristics, be computationally efficient, and also provide a smooth transition between the pre-sliding and sliding regimes. Various friction models proposed in the literature describe friction force/torque in terms of dynamic friction phenomena, like break-away force, presliding behavior, and frictional lag. These models include Dahl [5], LuGre [6], Leuven integrated friction model [7], Generalized Maxwell slip model (GMS) [8], Differential Algebraic Multistate (DAM) friction model [9], and smooth Generalized Maxwell slip model (S-GMS) [10], [11]. The Dahl and LuGre models are quite simple but they do not capture realistic friction phenomena like the Stribeck effect and hysteresis with non-local memory, and they can cause positional drift. The Leuven model exhibits hysteresis with non-local memory but it is difficult to implement. GMS and DAM capture all essential friction characteristics in pre-sliding and sliding motion. The GMS model presents a switching state condition between pre-sliding and sliding, which significantly complicates its application in real-time control. The DAM friction model has a similar structure as GMS, but without a switching condition. This model is not suitable for online parameter estimation because of its saturation function. Further, to overcome most of these limitations, Boegli et al. [11] recently developed a smoothed (analytic) version of the multi-state GMS model, referred to as the S-GMS friction model, which has an analytic set of differential equations highly suited for online parameter estimation.

Most model-based estimation used for online parameter estimation exploit Kalman filter (KF) techniques. A Kalman filter is an optimal state estimator for linear system since it results in minimization of the trace of the estimation error's covariance matrix. However, for nonlinear systems the generalization of the Kalman filter is formulated in terms of the Extended Kalman filter (EKF). The Extended Kalman filter is based on a linearization of the systems' dynamics using a first order Taylor expansion. However, the local linearization assumption may negatively affect the accuracy of the state estimation.

The Iterated Extended Kalman filter (IEKF) [12] improves the state and parameter convergence compared with the traditional EKF by iterating the state and parameter estimates more than once during each sample period. The dynamics of many engineering systems can only be predicted with continuous-time dynamics, whereas the measurements are obtained at discrete instances of time. For such type of applications, the Hybrid Extended Kalman filter (HEK) [12] is proposed. To overcome the EKF flaws, the Unscented Kalman filter [13] may be also considered. UKF is a derivative-free state estimation method, that is as accurate as the Extended Kalman filter (EKF) since the UKF captures the mean and covariance up to second order, and, compared to Monte Carlo type of methods, the UKF uses deterministic sampling [13].

The rest of the paper is organized as follows. Section 3 presents the system model, Section 4 presents the selected friction model for model-based compensation, Section 5 presents estimation techniques used, Section 6 presents

friction compensation, and finally Sections 7 and 8 discusses and concludes the paper.

3 SYSTEM MODEL

In order to evaluate the performance of each estimation technique, the actuation system of a 6-DOF haptic device is considered. The input torque applied to the shaft to control the angular position q of the rotating mass with the inertia J . The friction of the system originates from the shaft bearing and the linear guideway.

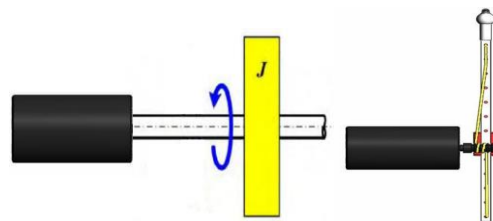


Fig. 1 Physical model of the system

The equation of motion for the system is given by

$$J\ddot{q} = u - F_f \quad (1)$$

Where u is the input torque, F_f is the friction force/torque originating from the shaft bearing and the linear guideway. The friction is estimated based on smoothed generalized Maxwell slip model (S-GMS).

4 FRICTION MODEL

In this paper the multi-state smoothed Generalized Maxwell slip model (S-GMS) [11] is used to estimate the friction in the physical model of the system shown in Fig. 1. S-GMS is a multi-state dynamic friction model that is an extension of the Generalized Maxwell-Slip (GMS) friction model [8]. This friction model implements all complex friction phenomena present in the GMS model, such as stick-slip motion, pre-sliding displacement, Stribeck effects, frictional lag, and transitions between sliding and pre-sliding regimes, while maintaining simplicity appropriate for control purposes. The S-GMS model consists of a set of differential equations in an analytical form, which makes it suitable for online friction estimation. Furthermore, S-GMS preserve the benefits of the GMS model by combining the state equations of pre-sliding and sliding into a single state equation for each element i . The analytical multi-state S-GMS model according to [11] is given in equation (2)

$$\frac{dF_i}{dt} = k_i(\dot{q} - \eta_i(F_i, \dot{q})|\dot{q}| \frac{F_i}{\alpha_{is}(\dot{q})}) \quad (2)$$

Where $\eta_i(F_i, \dot{q})$ is a smoothed transition function between the pre-sliding ($\eta_i = 0$) and sliding ($\eta_i = 1$) regimes for each element i . The smooth transition function $\eta_i(F_i, \dot{q})$ consists of two multiplicative terms given in equation (3), the transition function (4) and the reset function (5).

$$\eta_i(F_i, \dot{q}) = \eta_{A_i}(F_i, \dot{q}) \eta_{B_i}(F_i, \dot{q}) \quad (3)$$

$$\eta_{A_i}(F_i, \dot{q}) = 1 - \frac{1}{2} \tanh \left[\lambda \left(\frac{F_i}{\alpha_i s(\dot{q})} + \varsigma \right) \right] + \frac{1}{2} \tanh \left[\lambda \left(\frac{F_i}{\alpha_i s(\dot{q})} - \varsigma \right) \right] \quad (4)$$

$$\eta_{B_i}(z_i, \dot{q}) = \frac{1}{2} + \frac{1}{2} \tanh \left[\gamma \left(\frac{F_i}{\alpha_i s(\dot{q})} \frac{\dot{q}}{v_s} \right) \right] \quad (5)$$

Where $\varsigma \in [0.9, 1]$ is a factor to avoid ringing in the sliding regime and to set the frictional lag. The factor λ sets the transition sharpness of (4). The reset function (5) vanishes when the velocity reverses. The factor γ sets the transition sharpness of (5), while α_i is the normalized sustainable maximum friction force of each element i during sticking defined in equation (6), and $s(\dot{q})$ is the Stribeck curve defined in equation (7).

$$\alpha_i = \frac{F_i}{\sum_{i=1}^n F_i} \quad (6)$$

$$s(\dot{q}) = F_c + (F_s - F_c) e^{-\left| \frac{\dot{q}}{v_s} \right|^2} \quad (7)$$

Where F_s and F_c are static and Coulomb friction forces, v_s the Stribeck velocity, k_i is the stiffness of element i . The resulting friction force F_f is the summation of n number of stiffness forces F_i plus a viscous $\sigma_v \dot{q}$ term is given in (8).

$$F_f = \sum_{i=1}^N F_i + \sigma_v \dot{q} \quad (8)$$

5 FRICTION ESTIMATION AND COMPENSATION

In this section the friction state estimation framework is presented. A brief description of the scrutinized filters is given in the following subsections. The estimation problem is based on a discrete version of equation (2) and (8), discretized at time instant t_k given in equation (9) and (10).

$$F_k = f(F_{k-1}, \dot{q}_{k-1}) + w_{k-1}, \quad w_k \sim N(0, Q_k) \quad (9)$$

$$F_{f_k} = h(F_k, \dot{q}_k) + v_k, \quad v_k \sim N(0, R_k) \quad (10)$$

Here F_k and F_{f_k} represent the state and measurement vectors at time instant t_k , while $f(\cdot)$ and $h(\cdot)$ are non-linear functions representing the process model (friction state equation (2)) and the observation model (total friction equation (8)). Where w_k and v_k are the process model noise and the observation model noise with covariance Q_k and R_k , respectively.

In order to compensate for friction, the observer-based friction compensation strategy proposed in [10]-[11], as shown in Fig. 2, has been used, where the objective is to estimate the friction state F_i of equation (9), while minimizing the error between estimated friction $\hat{F}_{f_k}$ (equation

(10) and the measured friction $\tilde{F}_{f_k}$, based on the control input u_k and the inertia J of the system.

The above estimation framework, which is based on the S-GMS model, is implemented with the Extended, Iterated Extended, hybrid Extended, and Unscented Kalman filters. A brief description of these filters is given in the following subsections.

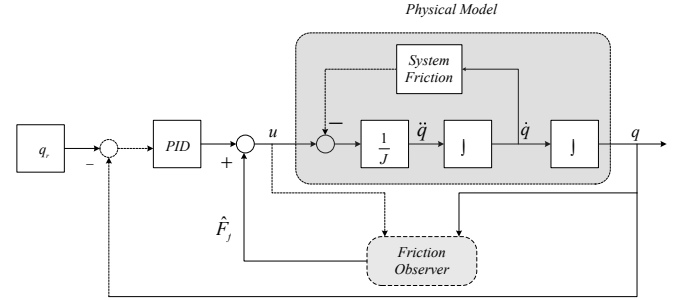


Fig. 2 Friction estimation and compensation

5.1 Extended Kalman filter (EKF)

The extended Kalman filter is a recursive filter that consists of time update and measurement update steps. It predicts the future state of $\hat{F}_k^-$ based on the available system model $f(\cdot)$ and projects ahead the state error covariance matrix P_k^- using the time update equations to compute the predicted measurement from the predicted state using function $h(\cdot)$. The predictor step (time update) equations are as follow,

$$\hat{F}_k^- = f(F_{k-1}, \dot{q}_{k-1}) \quad (11)$$

$$P_k^- = A_{k-1} P_{k-1}^+ A_{k-1}^T + Q_{k-1} \quad (12)$$

The corrector step (measurement-update equations) incorporates the measurement $\tilde{F}_{f_k}$ into an estimated state $\hat{F}_{f_k}$ using the Kalman gain matrix K_k . The state error covariance for the updated state estimate P_k is also computed using the following measurement update equations:

$$K_k = P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} \quad (13)$$

$$\hat{F}_k^+ = \hat{F}_k^- + K_k (\tilde{F}_{f_k} - h(\hat{F}_k^-, \dot{q}_k)) \quad (14)$$

$$P_k^+ = (I - K_k H_k) P_k^- \quad (15)$$

Where I is an identity matrix, while the state transition matrix A_k and observation matrix H_k are defined by first order Taylor series approximation of friction state equation (2), and observation equation (8) using the following Jacobian matrices.

$$A_{k-1} = \left. \frac{\partial f}{\partial F} \right|_{\hat{F}_{k-1}} \quad \text{and} \quad H_k = \left. \frac{\partial h}{\partial F} \right|_{\hat{F}_{k-1}}$$

5.2 Iterated Extended Kalman filter (IEKF)

The EKF linearize the state and measurement equations about the predicted state at an operating point. This prediction is often inaccurate in practice. The IEKF reduces the

linearization error in EKF. The estimate can be refined by re-evaluating the filter around the new estimated state operating point. This refinement procedure can be iterated until little extra improvement is obtained. The IEKF also consists of time update and measurement update steps like the EKF. The predictor step (time update) is the same as the EKF time update step given in equation (11) and (12).

The corrector step (measurement update) is performed by initializing the iterated EKF estimate to the standard EKF estimate:

$$\begin{aligned}\hat{F}_{k,0}^+ &= \hat{F}_k^- \\ P_{k,0}^+ &= P_k^-\end{aligned}\quad (16)$$

For $i = 0, 1, \dots, N$, evaluate the following equations (where N is the desired number of measurement-update iterations) [12]:

$$K_{k,i} = P_k^- H_{k,i}^T (H_{k,i} P_k^- H_{k,i}^T + R_k)^{-1} \quad (17)$$

$$P_{k,i+1}^+ = (I - K_{k,i} H_{k,i}) P_k^- \quad (18)$$

$$\hat{F}_{k,i+1}^+ = \hat{F}_k^- + K_{k,i} [\tilde{F}_{f_k} - h(\hat{F}_k^-, \dot{q}_k) - H_{k,i} (\hat{F}_k^- - \hat{F}_{k,i}^+)] \quad (19)$$

The final a posteriori state estimate and estimation error covariance are as follows:

$$\begin{aligned}\hat{F}_k^+ &= \hat{F}_{k,N+1}^+ \\ P_k^+ &= P_{k,N+1}^+\end{aligned}\quad (20)$$

5.3 Hybrid Extended Kalman filter (HEKF)

The Hybrid Extended Kalman filter (HEKF) treats the system as a continuous-time dynamic system but use a discrete-time measurement signal. The algorithm is briefly described below.

Initially the system equations with continuous-time dynamics and discrete-time measurements are defined as in equation (9) and (10).

For $k = 0, 1, \dots$, perform the following steps

- (a) Integrate the state estimate and its covariance from time $(k-1)$ to time k

$$\dot{\hat{F}}_k^- = f(F_{k-1}^-, \dot{q}_{k-1}) \quad (21)$$

$$\dot{P}_k^- = A_{k-1}^- P_{k-1}^- A_{k-1}^{T^-} + Q_{k-1}^- \quad (22)$$

The integration process begins with $\hat{F}_k^- = \hat{F}_{k-1}^-$ and $P = P_{k-1}^-$.

- (b). At time k , incorporate the measurement $\tilde{F}_{f_k}$ into the state estimate and estimation covariance as follow

$$K_k = P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} \quad (23)$$

$$\hat{F}_{k,i+1}^+ = \hat{F}_k^- + K_k [\tilde{F}_{f_k} - h(\hat{F}_k^-, \dot{q}_k)] \quad (24)$$

$$P_k^+ = (I - K_k H_k) P_k^- (I - K_k H_k)^T + K_k R_k K_k^T \quad (25)$$

5.4 Unscented Kalman filter

The UKF performs stochastic linearization through the use of a weighted statistical linear regression process. The UKF uses *unscented transform* (UT) as linearization method instead of Taylor series expansion. The UKF

deterministically extract the so called *sigma points* from the Gaussian and passes these through a function g . These *sigma points* are located at the mean and symmetrically along the main axis of the covariance and capture the true mean and covariance of the Gaussian random variable, and when propagated through the true non-linear system, captures the posteriori mean and covariance accurately to the 3rd order (Taylor series expansion) for any nonlinearity. Let us as an example propagate a random variable x (n -dimension) through a nonlinear function $y = g(x)$ with mean $\bar{x}$ and covariance P_x . To calculate the statistics of y , we form a matrix χ of $2n+1$ *sigma points* χ_i with corresponding weights W_i , where the *sigma points* are chosen according to the following rule:

$$\chi_o = \bar{x} \quad (26)$$

$$\chi_i = \bar{x} + \left(\sqrt{(n+\lambda)P_x} \right)_i, \text{ for } i = 1, \dots, n \quad (27)$$

$$\chi_i = \bar{x} - \left(\sqrt{(n+\lambda)P_x} \right)_{i-n}, \text{ for } i = n+1, \dots, 2n \quad (28)$$

$$W_i^{(c)} = \frac{\lambda}{\lambda+1} + (1-\alpha^2 + \beta) \quad (29)$$

$$W_i^{(m)} = W_i^{(c)} = \frac{1}{2(\lambda+1)}, \text{ for } i = 1, \dots, 2n \quad (30)$$

Where $\lambda = \alpha^2(n+k) - n$ is a scaling parameter, α determines the spread of the sigma points around the mean $\bar{x}$ and is usually set to a small positive value of $1e-3$. k is a secondary scaling parameter which is usually set to 0, and β is used to incorporate prior knowledge of the distribution (for Gaussian distributions, $\beta = 2$ is optimal). $\sqrt{(n+\lambda)P_x}$ is the i^{th} row of the matrix square root. These sigma vectors are propagated through the nonlinear function,

$$\bar{y} = \sum_{i=0}^{2n} W_i^{(m)} Y_i \quad (31)$$

$$P_y \approx \sum_{i=0}^{2L} W_i^{(c)} \{Y_i - \bar{y}\} \{Y_i - \bar{y}\}^T \quad (32)$$

The UKF algorithm can be summarized as follow

1. Mean and covariance initialization
2. Calculate sigma points using equation (26), (27) and (28).

$$\chi_i = [\bar{x} \quad \bar{x} + \sqrt{(n+\lambda)P_x} \quad \bar{x} - \sqrt{(n+\lambda)P_x}]$$

3. Time update from x_{k-1}, P_{k-1} to x, P using unscented transform.

$$\chi_{k|k-i}^x = f[\chi_{k-1}^x, \chi_{k-1}^y]$$

$$\hat{x}_k^- = \sum_{i=0}^{2n} W_i^{(m)} \chi_{i,k|k-i}^x$$

$$P_k^i = \sum_{i=0}^{2n} W_i^{(c)} [\chi_{i,k|k-i}^x \quad \hat{x}_k^-] [\chi_{i,k|k-i}^x \quad \hat{x}_k^-]^T$$

$$y_{k|k-1} = H[\chi_{i,k|k-i}^x, \chi_{k-1}^n]$$

$$\hat{y}_k^- = \sum_{i=0}^{2n} W_i^{(m)} y_{i,k|k-i}^x$$

4. Measurement update

$$P\tilde{y}_k\tilde{y}_k = \sum_{i=0}^{2n} W_i^{(c)} \begin{bmatrix} y_{i,k|k-i}^x & \hat{y}_k^- \end{bmatrix} \begin{bmatrix} y_{i,k|k-i}^x & \hat{y}_k^- \end{bmatrix}^T$$

$$Px_k y_k = \sum_{i=0}^{2n} W_i^{(c)} \begin{bmatrix} \chi_{i,k|k-i}^x & \hat{x}_k^- \end{bmatrix} \begin{bmatrix} y_{i,k|k-i}^x & \hat{y}_k^- \end{bmatrix}^T$$

$$K = Px_k y_k P_{\tilde{y}_k\tilde{y}_k}^{-1}$$

$$\hat{x}_k = \hat{x}_k^- + K(y_k - \hat{y}_k^-)$$

$$P_k = P_k^- - KP_{\tilde{y}_k\tilde{y}_k}K^T$$

6 NUMERICAL EVALUATION AND EXPERIMENTAL VALIDATION

In order to evaluate the computational and estimation capability of each estimation technique, the estimation scheme shown in Fig. 2 is applied on the actuation system of the Ares haptic device [2]. The S-GMS friction model with 4 elementary blocks is sufficient for modeling the real frictional behavior. The nominal parameter values in the S-GMS model used in the friction observer are given in Table 1. These nominal parameter values are identified from the experimental actuator input output data for the single actuating system of a 6-DOF haptic device, shown in Fig. 1.

Table 1 Friction model parameters

Pre-sliding region parameter			Sliding region parameter		
Parameter	Value	unit	Parameter	Value	unit
k_1	1×10^4	$\frac{\text{Nm}}{\text{rad}}$	F_s	0.335	Nm
k_2	1×10^4	$\frac{\text{Nm}}{\text{rad}}$	F_c	0.285	Nm
k_3	1×10^4	$\frac{\text{Nm}}{\text{rad}}$	v_s	0.01	$\frac{\text{rad}}{\text{s}}$
k_4	1×10^4	$\frac{\text{Nm}}{\text{rad}}$	σ_v	0.018	$\frac{\text{Ns}}{\text{rad}}$
-	-	-	$\alpha_1, \alpha_2, \alpha_3, \alpha_4$	0.25	-
-	-	-	λ	1000	-
-	-	-	γ	100	-
-	-	-	ζ	0.97	-

The estimation techniques were analyzed both offline and online.

6.1 Offline estimation

First an offline scheme was used to evaluate the estimation techniques. For numerical evaluation a sine reference trajectory with amplitude $200\mu\text{m}$ and a frequency of 5Hz along a PID controller was applied to control the position of the system. The gains of the PID controller was chosen to be $K_p = 2 \frac{\text{Nm}}{\text{rad}}$, $K_i = 2 \frac{\text{Nm}}{\text{rads}}$ and $K_d = 100 \frac{\text{Nms}}{\text{rad}}$ with a sampling time of 1ms. In offline, first an estimated velocity was

obtained by taking the finite difference derivative of the measured position, sampled at 1ms, combined with a low pass filter with a cutoff frequency of 50 Hz. The measured friction was estimated based on the relation between control input u and inertial forces $J\ddot{q}$ of equation (1) as $\tilde{F}_f = u - J\ddot{q}$.

The four estimation techniques EKF, IEKF, HEKF and UKF were evaluated and compared based on computational efficiency, normalized root-mean square error (NRMSE), and standard deviation between measured and estimated friction over a period of the sine trajectory. The simulation results are summarized in Table 2, and a comparison between measured and estimated friction are shown in Fig. 3, Fig. 4, Fig. 5, and Fig. 6 for EKF, IEKF, HEKF and UKF, respectively. The numerical evaluation of the results in Table 2 shows that IEKF gives better results in terms of standard deviation and NRMSE, as compared to the other studied estimation techniques. UKF also gives good results but with a significantly high computational cost. The IEKF was simulated for one measurement-update iteration. Increasing the number of measurement-update iterations increased the computational cost but did not significantly improve the performance.

Table 2 Comparison between EKF, IEKF, HEKF and UKF

Comparison	Standard deviation	NRMSE[%]	Time[s]
EKF	3.2e-3	1.19	0.5066
IEKF	9.45e-6	7.6e-3	0.5871
HEKF	1.53e-2	4.74	0.8631
UKF	6.4e-4	2.4e-1	13.589

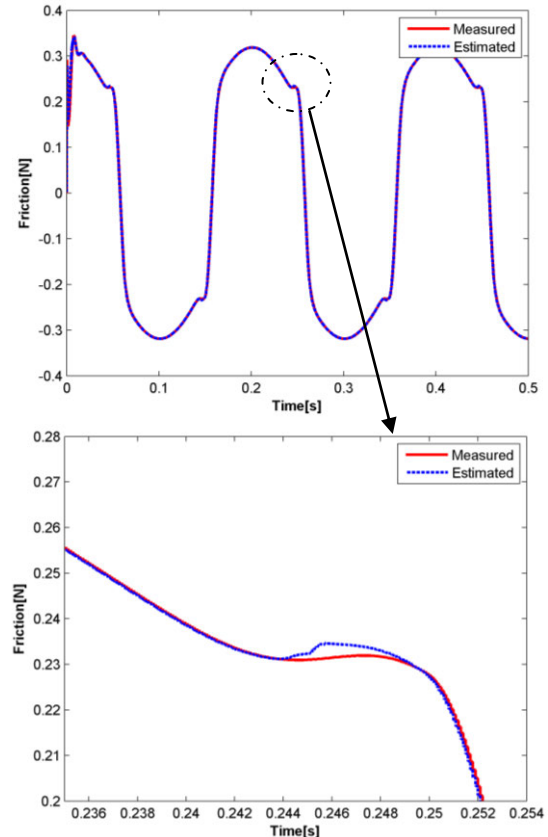


Fig. 3 Comparison of measured and estimated friction using EKF

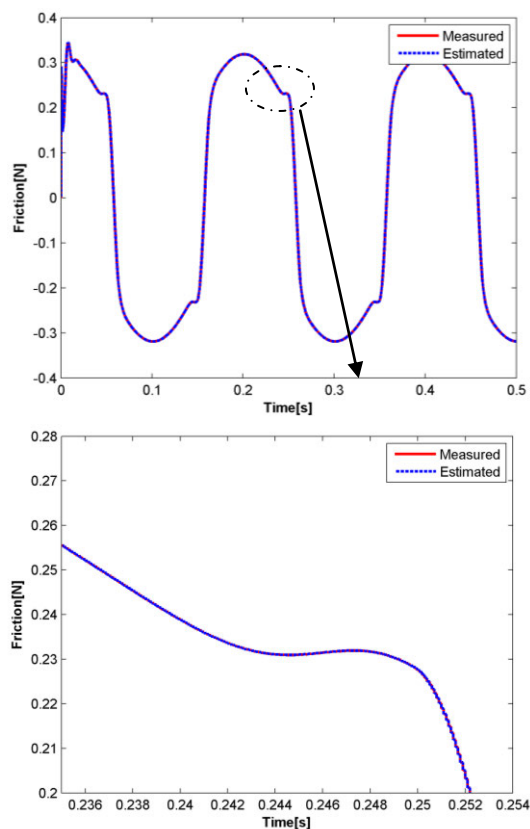


Fig. 4 Comparison of measured and estimated position using IEKF

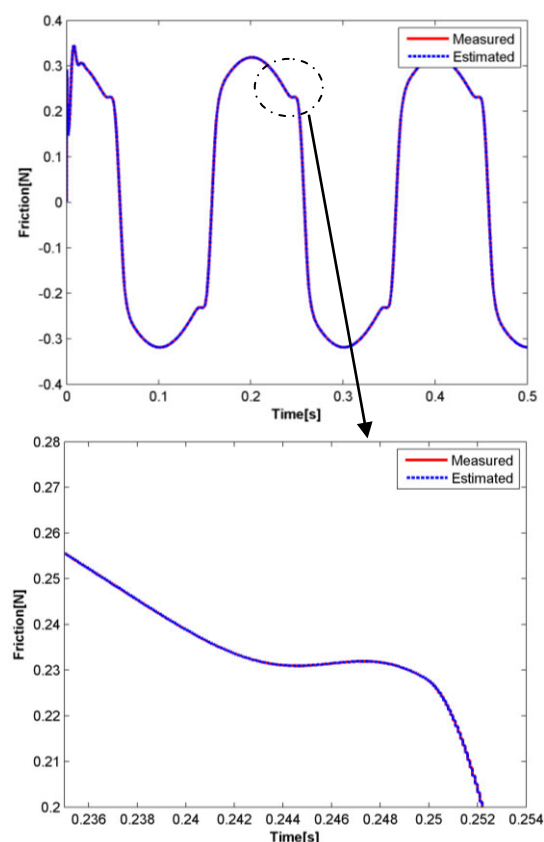


Fig. 6 Comparison of measured and estimated position using UKF

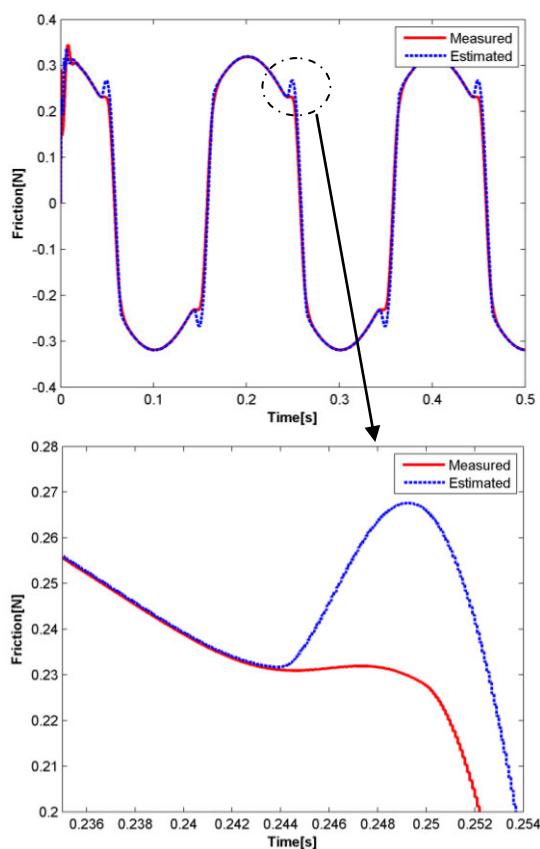


Fig. 5 Comparison of measured and estimated position using HEKF

6.2 Online experimental validation

In order to evaluate the performance capability of the estimation techniques for online friction estimation, again the actuation system of a 6-DOF haptic device based on a Stewart-platform was used. The experimental setup is shown in Fig. 7.

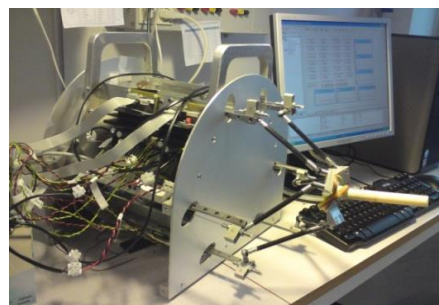


Fig. 7 Experimental setup

For experimental validation, the same sine wave reference trajectory with amplitude of $200\mu\text{m}$ and a frequency of 5Hz along a PID controller was used to control the position of the linear guideway. The same gains PID controller and sampling time was used as in the offline estimation. The estimation scheme shown in Fig. 2 was used to estimate and compensate friction with the same technique for velocity estimation and friction measurement.

The evaluation criteria for the online experimental validation are based on the normalized root-mean square error (NRMSE) and the standard deviation between measured and estimated friction. The results from the experiments are summarized in Table 3, while comparisons between measured

and estimated friction are shown in Fig. 8, Fig. 9, Fig. 10, and Fig. 11 for EKF, IEKF, HEKF and UKF, respectively. Table 3 shows that IEKF gives better results in terms of standard deviation and NRMSE as compared to the other estimation techniques. As in the offline estimation experiment, IEKF was simulated for one measurement-update iteration. Table 3 also confirms the off-line estimation results in Table 2.

Table 3 Experimental evaluation between EKF, IEKF, HEKF and UKF

Comparison	Standard deviation	NRMSE[%]
EKF	1.9e-3	1.4784
IEKF	2.67e-6	0.064
HEKF	3.2e-3	0.8783
UKF	1.5e-3	1.8827

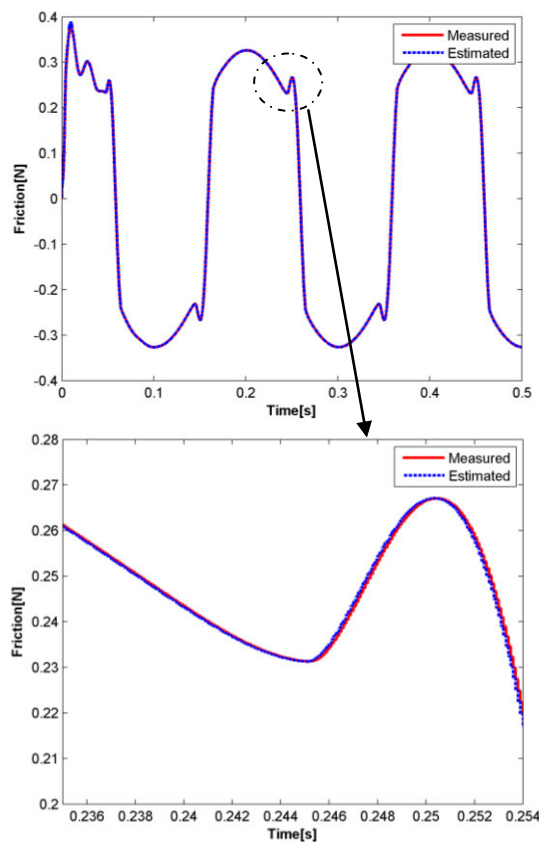


Fig. 8 Comparison of measured and estimated friction using EKF

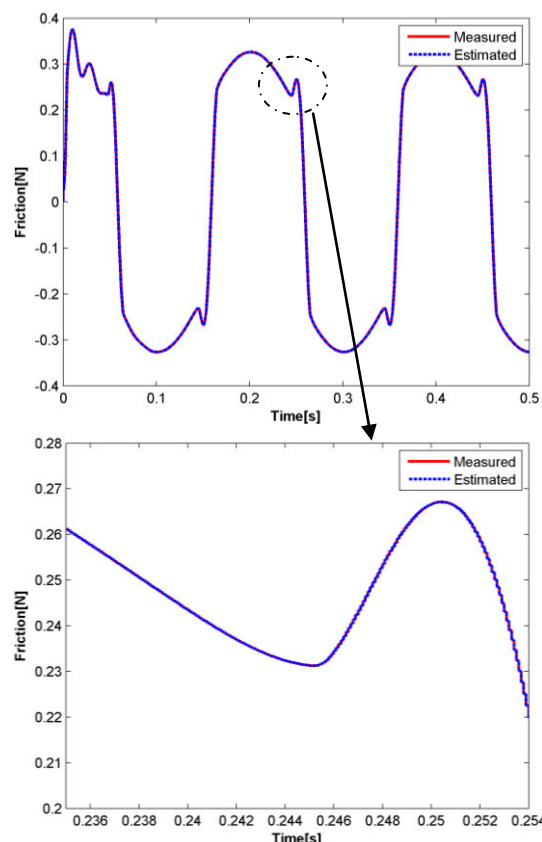


Fig. 9 Comparison of measured and estimated position using IEKF

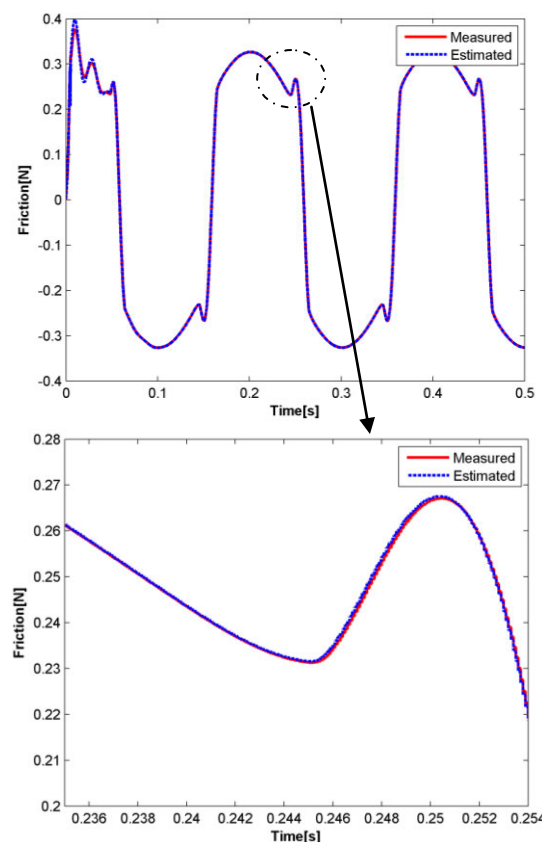


Fig. 10 Comparison of measured and estimated position using HEKF

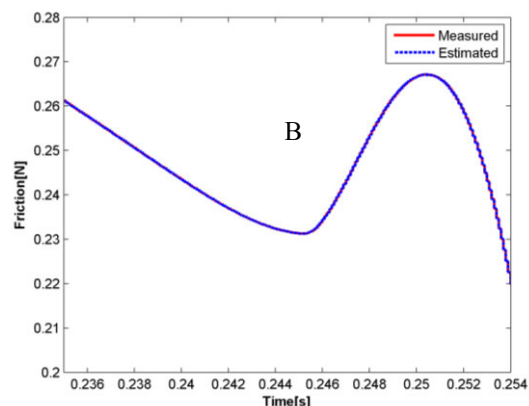
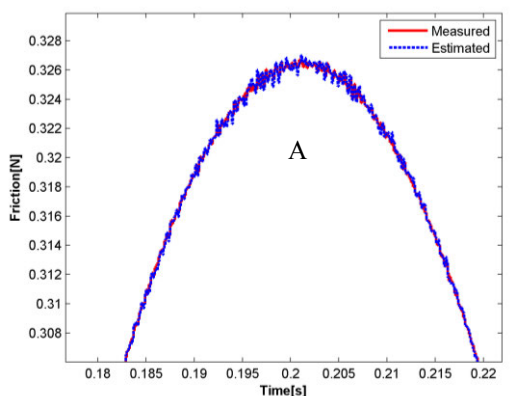
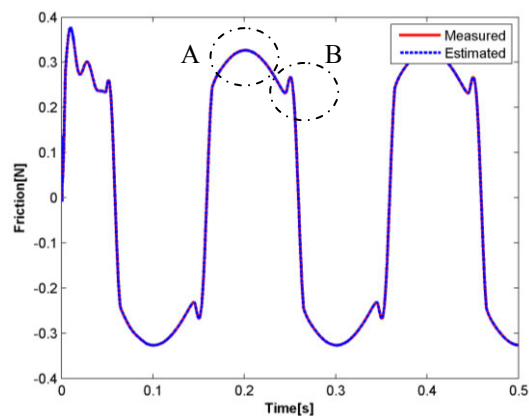


Fig. 11 Comparison of measured and estimated position using UKF

6.3 An experimental comparative analysis

An experimental comparative analysis of the system without friction compensation and with friction compensation based on the four estimation techniques was made. A comparison of the reference and measured positions for the case without compensation is shown in Fig. 12, while the cases with friction compensation are shown in Fig. 13. The tracking errors without compensation and with estimation and compensation using EKF, IEKF, HEKF and UKF are presented in Table 4.

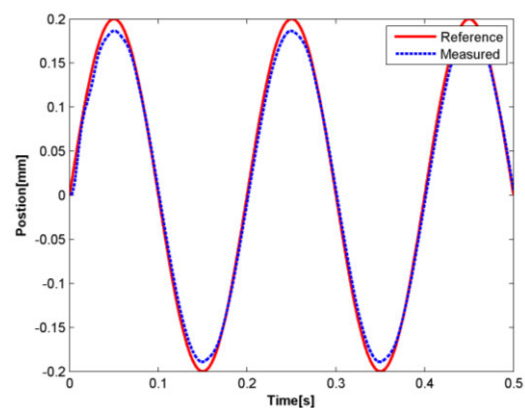


Fig. 12 Without friction compensation

Table 4 Position tracking error

Comparison		Standard deviation	NRMSE[%]
Without compensation		9.1e-2	6.82
With compensation	EKF	6.1e-3	3.66
	IEKF	5.1e-5	3.61
	HEKF	3.1e-3	3.69
	UKF	3.6e-3	3.68

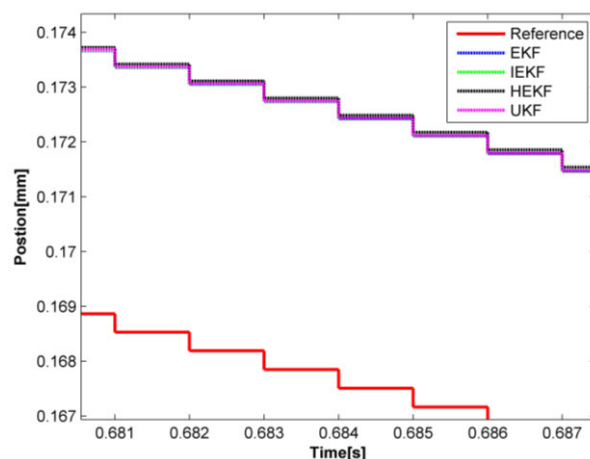
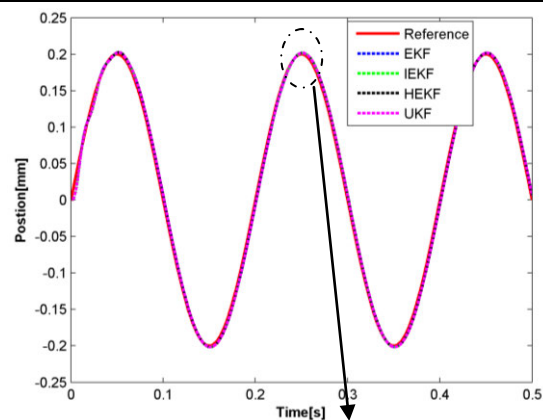


Fig. 13 Comparison of EKF, IEKF, HEKF and UKF with friction compensation.

7 DISCUSSION

In this comparative study we have exploited different estimation techniques based on the Kalman filter, such as Extended Kalman filter (EKF), Iterated Extended Kalman filter (IEKF), Hybrid Extended Kalman filter (HEKF), and

Unscented Kalman filter (UKF). Both online and offline strategies were used to evaluate and compare their performance based on standard deviation, normalized root mean-square error (NRMSE), and computational efficiency. The experimental evaluation (Table 2) confirms the low standard deviation for the Iterated Extended Kalman filter (IEKF) obtained in the offline estimation (Table 3). The online NRMSE is higher than the offline estimated but it is still the lowest among the tested filters. Further, the system was analyzed both with and without friction compensation in order to evaluate the effectiveness of each approach. A comparison with and without friction compensation (shown in Table 4), clearly indicates that position tracking without friction compensation results in a large standard deviation and NRMSE between the reference and measured trajectories. The resulting standard deviation of EKF, IEKF, HEKF and UKF are equal, while the IEKF has reduced NRMSE for friction estimation.

8 CONCLUSIONS

Transparency is a key performance evaluation criterion for haptic devices, which describes how realistically the haptic force/torque feedback is mimicked from a virtual environment. To achieve a good transparency for haptic devices we must be able to accurately estimate and compensate for disturbance forces like friction between moving parts. An accurate estimate of friction forces for observer based compensation requires computationally efficient estimation techniques that significantly reduce the error between measured and estimated friction. In this work, estimation techniques based on Kalman filter, such as Extended Kalman filter (EKF), Iterated Extended Kalman filter (IEKF), Hybrid extended Kalman filter (HEKF) and Unscented Kalman filter (UKF), have been implemented and compared with the aim to determine which estimation technique that is most efficient for friction compensation using online estimation. An observer-based friction compensation scheme was adopted using a model-based compensation scheme using the mentioned estimation techniques. The friction observer is based on a newly developed friction smooth generalized Maxwell slip model (S-GMS), which is suitable for online parameter estimation due to its smooth transition between the pre-sliding and sliding regimes. Each estimation technique was evaluated by numerical and experimental tests of sinusoidal position tracking. The performances of the system were evaluated for the standard deviation, normalized root mean-square error (NRMSE), and the computational time. The results indicate that friction estimation and compensation based on an Iterated Extended Kalman filter has the best performance, both in terms of accuracy (standard deviation and normalized root mean-square error (NRMSE) between measured and estimated friction) and computational efficiency. Future research will focus on friction estimation and compensation using the S-GMS friction model along with IEKF for complete 6-DOF motion control of the Ares haptic device.

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